

## Problem

Design a more complex problem than given in Prob. 14.10, to help other students better understand how to determine the Bode magnitude and phase plots of a given transfer function in terms of  $j\omega$ . Include at least a second order repeated root.

## Step-by-step solution

## Step 1 of 1

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Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

**Problem**

Sketch the magnitude phase Bode plot for the transfer function

$$H(\omega) = \frac{25j\omega}{(j\omega + 1)(j\omega + 5)^2(j\omega + 10)}$$

**Solution**

$$H(\omega) = \frac{\left(\frac{1}{100}\right)j\omega}{(1 + j\omega)\left(1 + \frac{j\omega}{5}\right)^2\left(1 + \frac{j\omega}{10}\right)}$$

$$20\log(1/100) = -40$$

For the plots, see the next page.